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April 14, 1976
SUPERSEDING
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FEDERAL SPECIFICATION

LEATHER, CATTLEHIDE, (BAG AND CASE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for vegetable tanned cattlehide leather (see 6.1).

1.2 Classification.

1.2.1 Types, classes, styles and forms. The leather shall be furnished in the following types, classes, styles and forms as specified (see 6.2).

Type I - Bag leather.
Type II - Case leather.
Class 1 - Mildew resistant treated.
Class 2 - Not mildew resistant treated.
Style A - Grain side smooth.
Style B - Boarded.
Style C - Grain side embossed in fine grain.
Style D - Grain side embossed in coarse grain.
Form 1 - Sides.
Form 2 - Bends.
Form 3 - Double shoulders.
Form 4 - Single shoulders.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Test Method Std. No. 311 - Leather, Methods of Sampling and Testing.

FSC 8330

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Military Specification:

MIL-L-43432 - Leather, Packaging and Packing of.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The leather shall be full grain or corrected grain, vegetable tanned cattlehide. Corrected grain leather shall be full grain leather that has been lightly snuffed on the grain side to remove superficial blemishes. The flesh side may be buffed, shaved or split to secure greater uniformity of thickness.

3.2 Physical requirements.

3.2.1 Trim (form and area). The leather shall be furnished as sides, bends, double shoulders or single shoulders as required for the form specified (see 1.2.1 and 6.2) and shall be trimmed to conform with the definition of these terms as listed under "Definitions of Terms" in FED-STD-311. The area of each side shall be not less than 18 square feet. The area expressed in terms of square feet shall be legibly marked on the flesh side of the leather. Sides and bends shall be marked in the butt area, shoulders shall be marked in the corner formed by the straight edge cut and the belly cut.

3.2.2 Thickness. The thickness of the leather shall be as specified (see 6.2 and 6.3). A tolerance of plus 1 ounce and minus 1/2 ounce shall be permitted. (NOTE: 1 ounce = 1/64 inch).

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3.2.3 Color and finish. The color and finish of the leather shall be as specified (see 6.2).

3.2.4 Styles.

3.2.4.1 Style A. The grain side of style A leather shall be smooth.

3.2.4.2 Style B. The grain side of style B leather shall have a boarded effect.

3.2.4.3 Style C. The grain side of style C leather shall be embossed in a fine grain.

3.2.4.4 Style D. The grain side of style D leather shall be embossed in a coarse grain.

3.2.5 Breaking force. At least 80 percent of the specimens tested shall meet the minimum breaking force specified in table I, when tested as specified in 4.3.

3.2.6 Elongation. At least 80 percent of the specimens tested shall have elongation values no greater than 25 percent at the applicable load specified in table I, when tested as specified in 4.3.

TABLE I. Breaking force and elongation (types I and II)

Thickness (ounces)	Breaking force, minimum (pounds)	Elongation load (pounds)
3, 4 and 5	50	25
6, 7 and 8	100	50

3.2.7 Colorfastness (resistance to rubbing). The leather shall be tested for resistance to dry and wet crocking in accordance with 4.3. Staining of the dry cloth shall be not lower than Munsell value 8.5 and staining of the wet cloth shall be not lower than Munsell value 6.5.

3.2.8 Cracking. The leather shall not crack when tested as specified in 4.3.

3.3 Chemical requirements.

3.3.1 The leather shall conform to the chemical requirements of table II when tested as specified in 4.3.

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TABLE II. Chemical requirements

Characteristics	Minimum	Maximum
Chloroform soluble material, percent 1/:		
Type I	6.0	15.0
Type II	3.0	9.0
Water soluble material, percent 1/	-	13.0
Total ash, percent 1/	-	5.0
Acidity (pH)	3.3	5.0

1/ Calculated on the moisture free basis.

3.3.2 Class 1. When class 1 is specified (see 6.2), the leather shall contain not less than 0.10 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.3.

3.4 Workmanship. The finished leather shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Sampling for inspection shall be performed in accordance with FED-STD-311, except where otherwise indicated hereinafter.

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4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The defects found during examination of the leather for visual and dimensional characteristics shall be in accordance with 4.2.2.1 and 4.2.2.2. The sample unit for these examinations shall be one unit of product. The lot size shall be expressed in units of one unit of product. Any sample unit containing one or more defects shall be classified as a defective unit. Sampling for these examinations shall be performed in accordance with FED-STD-311, except that a sample size of 15 and a rejection number of 2 shall be applicable for thickness determinations on lots consisting of 51 or more units of product. Separate examinations shall be performed for visual and dimensional examinations, but the same sample may be used for each examination.

4.2.2.1 Examination for visual characteristics.

Examine	Defect
Form	Not full sides, bends, double shoulders or single shoulders as required for the form specified.
Color	Not as specified.
Style	Not smooth, boarded or embossed as specified.
Quality of leather	Grain not as specified. Flesh side not smooth or has areas of loose flesh in excess of four square inches in total area. Stain, burn mark, cut, scratch, hole, brand, hip mark, rough shoulder, heavy vein, deep fat wrinkle, boney or flanky areas, or other surface imperfections which impair the cutting value of more than 12 percent of the total area of the leather.

NOTE: Light, well healed scratches and grub holes, light fat wrinkles, and slight stains shall not be classified as defects.

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4.2.2.2 Examination for dimensional characteristics.

Examine	Defect
Area (expressed in terms of square feet)	Area is less than area marked on leather. Area of side is less than 18 square feet.
Marking	Omitted, illegible or not located in area specified.
Thickness 1/	Not as specified.

1/ Thickness shall be measured in accordance with method 1011 of FED-STD-311. No individual thickness measurement shall be less than the minimum or greater than the maximum specified.

4.2.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-L-43432 to determine whether packaging, packing and marking comply with the section 5 requirements.

4.3 End item testing. The methods of testing specified in FED-STD-311, wherever applicable and as listed in table III, shall be followed. Sampling procedure and location from which the sample unit is to be obtained shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be rejected if any one of the following conditions exist:

a. More than three test failures occur for breaking force or elongation.

b. More than one test failure occurs for the remaining requirements applicable to the sample unit.

c. Any composite fails to meet the specified requirement.

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TABLE III. Tests

Characteristic	Requirement paragraph	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Composite
Material identification	3.1	1/	-	-
Breaking force	3.2.5	2021	X	-
Elongation	3.2.6	2021	X	-
Colorfastness (resistance to rubbing)	3.2.7	3031	X	-
Cracking	3.2.8	4.3.1	X	-
Chloroform-soluble material	3.3.1	6341	-	X
Water-soluble material	3.3.1	6341	-	X
Total ash	3.3.1	6421	-	X
Acidity (pH)	3.3.1	6621	-	X
Paranitrophenol	3.3.2	6711	-	X

1/ Unless otherwise specified, a certificate of compliance shall be submitted and will be acceptable for the stated requirement.

4.3.1 Cracking. The leather shall be tested for cracking in accordance with method 4011 of FED-STD-311. The diameters of the mandrels used for testing the leather shall be as follows:

Thickness of leather (ounces)	Diameter of mandrel (inch)
3 to 5	1/4
6 to 8	3/8

5. PREPARATION FOR DELIVERY

5.1 Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Leather shall be packaged in accordance with the applicable requirements of MIL-L-43432.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B and C. Leather shall be packed in accordance with the applicable requirements of MIL-L-43432.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123.

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5.3.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-L-43432.

6. NOTES

6.1 Intended use. The leather is intended primarily for use in Post Office delivery bags, money bags, handbags, slide-fastener envelopes, and physicians satchels.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, class, style and form required (see 1.2.1 and 6.4).
(c) Thickness required (see 3.2.2).
(d) Color and finish required (see 3.2.3).
(e) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).
(f) Marking required (see 5.3).

6.3 Specifications referencing this document should specify thickness in whole ounces and adhere to the thickness tolerances specified herein.

6.4 Form. When selecting the form of the leather to be specified, consideration should be given to the dimensions of the components to be cut from the leather.

Custodians:

Army - GL
Navy - SA
Air Force - 82

Review activities:

Army - MD
Navy - MC
DSA-CT

Preparing activity:

Army - GL

Civil Agency Coordinating Activities:

AGR
DC
GSA-FSS
INT
PO
TR
VA-DMS

Project No. 8330-0140

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